Kartik Mangal

Software Engineer II · Unity Game Developer · Multiplayer & Gameplay Systems

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SUMMARY

Unity game developer with 6 years of experience across multiplayer, real-money slot and mobile titles. Builds network-synced gameplay with Photon Fusion, physics-based mechanics, and editor tooling, with a focus on SOLID architecture, performance and AI-assisted workflows.

EXPERIENCE

Software Engineer II, Liquidnitro Games — Jun 2025 – Present
Unity 3D · C# · Photon Fusion · Perforce · Visual Studio

Multiplayer Mini Games

- Designed and implemented network-synced gameplay mechanics using Photon Fusion, keeping dynamic player and environment interactions in sync across all clients.
- Developed a dynamic path generation system and an obstacle hit-detection and avoidance system that adapts in real time to gameplay, for both player and AI navigation.
- Worked closely with designers and artists to optimise the performance and latency of multiplayer interactions.

Game Developer, Arrise Solutions Pvt. Ltd. — Jun 2023 – May 2025
Unity 3D · C# · SVN · Jenkins · Spine · NGUI · UGUI

Bespoke Slot Games

- Developed three categories of real-money slot games for web and mobile.
- Integrated art, multi-lingual fonts and animation using NGUI and the Spine controller.
- Implemented robust gameplay features in C# and Unity following SOLID principles, OOP and design patterns.
- Carried out QA testing, debugging and bug-fixing.
- Optimised assets for performance and compatibility across a wide range of devices.

Unity Developer, BR Softech — Mar 2023 – May 2023
Unity 3D · C# · SignalR · GitHub Desktop

- Unity development with SignalR-based real-time networking.

Game Developer, Totality Corp — Oct 2020 – Dec 2022
Unity 3D · C# · Photon PUN · Roblox · Lua

Vijayi Dash · Oct 2022 – Dec 2022

- Developed a complete obstacle system using ScriptableObjects and DOTween.
- Used the State design pattern and the New Input System to handle player locomotion.
- Used the Observer pattern to avoid runtime exceptions and unnecessary per-tick memory allocations.
- Implemented physics-based collision detection with primitive colliders and Rigidbody components.
- Optimised performance using profiling-led techniques.

Rann Bhumi · Nov 2021 – Sep 2022

- Developed a deathmatch combat game with five character classes, each with its own weapons and abilities.
- Built physics-based locomotion, modular weapon systems, UI screens and dynamic camera behaviours.
- Integrated PvP networking, host migration, RPCs for client- and server-specific logic, and game state synchronisation.

INDEPENDENT PROJECTS: BREADCRUMBS

Fowl Play Jul 2026 – Sep 2026

Unity 6 · URP 17 · C# · Cinemachine · New Input System · ScriptableObjects · Custom shaders

- Engineered physics-based third-person platforming (coyote time, air control, wall-slide, jelly squash and stretch), tuned entirely through ScriptableObjects.
- Built a pooled weapon and fire-mode framework and an Egg Launcher with momentum and recoil, an aim arc with a live preview and landing ring, and ballistics that land on the crosshair.
- Made the world paintable. Shots paint the exact surface they hit, building on the paint-as-traversal idea.

Blockout Studio Sep 2026

Unity Editor tooling · UPM package · C# · Python · pywebview · Preact · HTTP/SSE bridge

- Built Blockout Studio, an embeddable Unity package that turns a reference image or text prompt into a dimensioned, playable greybox level.
- Built a Python and Preact desktop app (pywebview) that talks to a running Unity Editor over a local HTTP/SSE bridge, with Blender sync for hand edits.
- Added a Theme Kit that restyles a whole level from a single reference image, and used it to build the "Water Gardens" demo level.

TECHNICAL SKILLS

Languages	C#, Lua, Python
Engines & Tools	Unity 6 / URP, Unity 3D/2D, Roblox Studio, Visual Studio, Blender (level bridge)
Gameplay & Graphics	Physics-based locomotion, Cinemachine, New Input System, Custom shaders & VFX, Spine, DOTween
Networking	Photon Fusion, Photon PUN, Netcode for GameObjects, SignalR
Architecture & Optimisation	SOLID, Design patterns, ScriptableObjects, Object pooling, Batching, Profiler, Debugging & testing
Tooling & AI	Unity Editor tooling, UPM packages, Claude Code (skills, agents, hooks), Unity CLI automation
UI	NGUI, UGUI / Canvas
Workflow	Perforce, Git / GitHub, SVN, Jenkins, JIRA, Monday

EDUCATION

Indian Institute of Information Technology, Jabalpur Jul 2018 – May 2022

B.Tech, Electronics and Communication Engineering · Jabalpur, India